HOW NOT TO ACQUIRE EXCHANGE RULES IN LOGICAL PHONOLOGY*

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1. Introduction

Since at least Meinhof (1912), phonologists have posited the existence of what have been called *exchange* (or *flip-flop* or *polarity*) processes, in which two phonological structures appear to be "exchanged" for each other, usually as an expression of a particular morphosyntactic category or context. For instance, if X and Y are phonological structures, a grammar is said to to exhibit an exchange if $X \rightsquigarrow Y$ and $Y \rightsquigarrow X$ in the same morphosyntactic context. For example, in San Martín Itunyoso Triqui (DiCanio et al. 2020), vowel length and the presence of a postvocalic [ɦ] "exchange" in the 3rd person singular proximate (3sg. prox.).

	inf.	3sg. prox.		
(1)	r**i**:^{32}	r**iɦ**^{3}	'take out'	V: ⇜⇝ ɦ
	a^{3}j**a**:^{32}	a^{3}j**aɦ**^{3}	'read'	
	a^{3}t**oɦ**^{3}	a^{3}**o**:^{3}	'laugh'	
	r**ãɦ**^{4}	r**ã**:^{3}	'buy'	

If we take the infinitive (inf.) to be the underlying form, then it appears that the 3sg. prox. is formed by replacing an underlying mora linked to the nucleus with postvocalic [ɦ], and underlying postvocalic [ɦ] with a mora linked to the nucleus. Many other putative exchanges, however, appear to involve single phonological features. A phonological grammar might be said to exhibit an exchange with respect to some feature F if

- $\begin{bmatrix} +\text{F} \\ \ldots \end{bmatrix} \rightsquigarrow \{-\text{F}\}$
- $\begin{bmatrix} -\text{F} \\ \ldots \end{bmatrix} \rightsquigarrow \{+\text{F}\}$

in the same morphosyntactic context. One such example from comes from Dholuo (Tucker 1994). It appears that the final oral obstruent in the noun stem takes on the opposing value of the feature Voice in the nominative singular (nom. sg.) and the plural (pl.).

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		nom. sg.	pl.		
(2)	a.	gɔt̠	gɔd̠ɛ	'hill'	t ↭ d
		kɛd̠ɪ	kɛt̠ɛ	'twig'	
	b.	agɔk̠ɔ	agɔg̠ɛ	'chest'	k ↭ g
		hɪg̠a	hik̠e	'year'	

Similar alternations apply to other oral obstruents and to some loanwords (e.g., [kitæb̠u]–[kitep̠e] 'book' from Arabic via Swahili). Other candidate exchanges have been proposed in Brussels Flemish Dutch (Zonneveld 1976), Czech (Wolfe 1970), English (Chomsky and Halle 1968: §4.4.3.8), and Menomini (Bever 1967: 86–94).

Exchange processes raise a number of questions for morphological theory in particular. Exchanges are usually morphosyntactically rather than phonologically conditioned; Anderson and Browne (1973) and de Lacy (2012) suggest this is a linguistic universal. At the same time, their phonological realization poses a challenge for piece-based (i.e., concatenative) theories of morphology. That is, it is not obvious how one might analyze the San Martín Itunyoso Triqui or Dholuo patterns in terms of affixation and phonological processes this affixation triggers, without reference to purely morphophonological *processes* of the very sort piece-based morphololgical theories tend to discount. However, we will not focus on these issues here.

We are skeptical as to whether exchange processes actually exist in the first place, and nearly all examples in the literature are controversial. For example, Moreton (2004) disputes Zonneveld's exchange analysis of Brussels Flemish, Anderson and Browne (1973) dispute Wolfe's exchange analysis of Czech, and McCawley (1974) criticizes Chomsky and Halle's use of exchanges in English.

De Lacy (2012) similarly disputes the traditional analysis of Dholuo plural formation (see citations in de Lacy 2012) as involving an exchange. Simplifying somewhat, he argues that the apparent exchange reflects distinct suffixes and phonological processes. De Lacy notes the existence of many non-alternating nouns, like [cup̠æ, cup̠e] 'bottle(s)' or [osik̠i, osik̠e] 'stump(s)', with a voiceless obstruent throughout the paradigm. In his analysis, all alternating (i.e., exchanging) nouns contain an underlying voiced obstruent. The plural suffix begins with an abstract segment which triggers vowel deletion and devoicing, as in /kɛd̠ɪ-Q̥E/ ⇝ [kɛt̠ɛ] 'twigs' (where /Q̥/ denotes this abstract segment and /E/ denotes an Atr-harmonic mid front vowel realized as [e, ɛ]). However, rather than taking the nom. sg. to be the citation form, he posits that it is derived from another abstract suffix, denoted /-X̠̥/, which triggers devoicing but does not delete a stem-final vowel, as in /gɔd̠-X̠̥/ ⇝ [gɔt̠] 'hills' or /agɔg̠ɔ-X̠̥/ ⇝ [agɔk̠ɔ] 'chests'. If this analysis is viable, then both 'twig' and 'hill' derive from underlying /d/ and there is no exchange to speak of.

Exchanges also pose potential problems for phonological theory, to which we now turn. De Lacy (2020) draws an important distinction between exchanges as descriptive, taxonomic objects and the actual theoretical mechanisms (e.g., "exchange rules") used to generate them. In classical generative phonology, the α-notation makes it trivial to write exchange rules, as shown in section 2. The problem is more complex in Logical Phonology

(LP; Bale et al. 2014; Bale and Reiss 2018; Bale et al. 2020; Reiss 2021; Dabbous et al. 2024; Leduc et al. 2024; Reiss 2024; Gorman and Reiss 2025a,b; Reiss this volume), a rule-based theory which proposes a novel set of phonological operators: LP lacks a mechanism to express exchange **rules**, but as shown in section 3, it can compute exchanges **processes** via a series of ordered rules. In section 4, we argue that this is not a problem and that the apparent non-existence of exchanges, phonologically or morphological, may reflect either the *diachronic filter* or non-trivial properties of the phonological *language acquisition device*.

2. Exchange rules in classical generative phonology

For sake of discussion—and in lieu of an uncontroversial, naturally-occurring example—we provide the following toy data set for "Fauxluo", loosely inspired by the traditional analysis of Dholuo.

(3)

	sg.	du.	tr.	pl.		
a.	nap	nap-ga	na<u>b</u>-lu	nap-a	'cactus'	p ⇜⇝ b
b.	nab	nab-ga	na<u>p</u>-lu	nab-a	'igloo'	

Fauxluo exhibits an exchange of Voice for labial-final stems in the trial (tr.), which is accompanied by the suffix /-lu/. For concreteness, we assume the exchange targets labial obstruents and is triggered by the +Lateral feature of the /l/ in the trial suffix, even though we note prior claims that all exchanges have morphosyntactic triggers.

In *The Sound Pattern of English* (SPE), Chomsky and Halle (1968: 355f.) take the existence of phonological exchanges for granted. They note that exchanges cannot be modeled with traditional phonological rules—i.e., without α-notation—because this produces an ordering paradox. As shown in (6), rules (4–5), applied in either order, do not yield the Fauxluo pattern.[1]

(4) $$\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \\ +\textsc{Voice} \end{bmatrix} \longrightarrow \{-\textsc{Voice}\} \;/\; \_\_ \; [+\textsc{Lateral}]$$

(5) $$\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \\ -\textsc{Voice} \end{bmatrix} \longrightarrow \{+\textsc{Voice}\} \;/\; \_\_ \; [+\textsc{Lateral}]$$

(6) Ordering paradox:

UR	nap-lu	nab-lu	UR	nap-lu	nab-lu
Rule (4):		na**p**lu	Rule (5):	na**b**lu	
Rule (5):	na**b**lu	na**b**lu	Rule (4):	na**p**lu	na**p**lu
SR	na**b**lu	*na**b**lu	SR	*na**p**lu	na**p**lu

[1]We adopt the anachronistic convention of writing the "change" portion of the rule using curly braces, a convention explained in section 3 below.

Intuitively, the problem here is that whichever rule applies first will neutralize the /p, b/ distinction which the other rule depends on. Chomsky and Halle, following a suggestion from Bever (1967), instead express exchanges using α-notation, permitting the Fauxluo exchange to be expressed as a single phonological rule which has the effect of applying (4–5) simultaneously.

(7) $$\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \\ \alpha\textsc{Voice} \end{bmatrix} \longrightarrow \{-\alpha\textsc{Voice}\} \; / \, \_\_ \; [+\textsc{Lateral}]$$

While SPE does not give a complete semantics for the α-notation (see Bale et al. 2025), the convention of negating α, as in the change $\{-\alpha\textsc{Voice}\}$, is also used to define rules of dissimilation (SPE:177f.). The sole novelty here is the αVOICE in the rule target, which serves to assign a value to the α-variable referred to in the change.

3. Exchange rules in Logical Phonology

Logical Phonology is also a rule-based theory, but it is distinguished by an insistence on a rigorous definition of natural classes and by the use of novel operators in lieu of the traditional $\rightarrow$. We first review these formal matters, then consider the status of exchange processes in LP.

3.1 Formal preliminaries

In LP, Universal Grammar is assumed to provide a universal and finite feature set, denoted $\mathcal{F}$ (e.g., Chomsky and Halle 1965; Reiss and Volenec 2022). Features are to be binary valued, so a *valued feature* is an element of $\{+,-\} \times \mathcal{F}$. A *feature specification* is thus a (possibly-empty) set of valued features. Feature specifications defining segments must be *consistent*: if a segment is specified +F, it cannot also be specified −F and vice versa. However, feature specifications need not be *complete*: underspecification is permitted. *Segments* are then (possibly-empty, consistent) feature specifications, linked to an X-slot. However, the X tier plays no role henceforth and thus will be omitted here. Three example segments are given below; note that /I/ is underspecified for HIGH.

(8) Segments as sets of valued features:

$$/\text{i}/ = \begin{Bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ +\textsc{High} \\ -\textsc{Low} \\ +\textsc{Atr} \end{Bmatrix} \qquad /\text{e}/ = \begin{Bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{High} \\ -\textsc{Low} \\ +\textsc{Atr} \end{Bmatrix} \qquad /\text{I}/ = \begin{Bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{Low} \\ +\textsc{Atr} \end{Bmatrix}$$

Natural classes are also defined by sets of valued features, but are interpreted as **sets of segments**: they match any segment whose specification is a *superset* of their specification

(Bale et al. 2020). To maintain type consistency, curly braces are thus used for segments and square brackets for natural classes.

(9) $\left[+\textsc{High}\right] = \left\{x \mid x \supseteq \left\{+\textsc{High}\right\}\right\} = \{\text{i}, \text{y}, \text{ɨ}, \text{ʉ}, \text{ɯ}, \text{u}, \ldots\}$

One consequence of this definition (e.g., Lees 1961:12f., Lightner 1971:236) is that one cannot specify a natural class matching an underspecified segment, say /I/, without also matching any segment whose specification is a proper superset of that segment, say, /i, e/. Below, we refer to this property, illustrated in (10), as *specificity*.

(10) $\begin{bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{Low} \\ +\textsc{Atr} \end{bmatrix} = \left\{x \mid x \supseteq \begin{Bmatrix} +\textsc{Syllabic} \\ -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{Low} \\ +\text{ATR} \end{Bmatrix}\right\} = \{\text{i}, \text{e}, \text{I}\}$

Another implication of this formulation is that the expression $[\ ]$ refers to the class of all segments, since every set of features is a superset of the empty set.

In lieu of $\rightarrow$, LP proposes three novel operators, two of which are relevant here.[2] The first is the *subtraction* operator denoted by $\setminus$. This operator has the same semantics as set difference and removes valued features from a segment.

(11) Subtraction: $A \setminus B = \{c\text{F} \mid c\text{F} \in A \wedge c\text{F} \notin B\}$

The second operator, the *unification* operator, is denoted by $\sqcup$. Adapted from the syntactic literature (e.g., Shieber 1986), it adds valued features to a segment, applying vacuously when the target is already specified for a given feature.

(12) Unification: $A \sqcup B = A \cup \{c\text{F} \mid c\text{F} \in B \wedge -c\text{F} \notin A\}$

In other words, $A \sqcup B$ is a segment containing all the valued features of A and any non-inconsistent features in B. This is exemplified with toy examples below.

(13) Yield of unification:

a. "Feature addition": $\begin{Bmatrix} -\textsc{Back} \\ -\textsc{Round} \end{Bmatrix} \sqcup \left\{-\textsc{High}\right\} \rightsquigarrow \begin{Bmatrix} -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{High} \end{Bmatrix}$

b. "Vacuous application": $\begin{Bmatrix} -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{High} \end{Bmatrix} \sqcup \left\{-\textsc{High}\right\} \rightsquigarrow \begin{Bmatrix} -\textsc{Back} \\ -\textsc{Round} \\ -\textsc{High} \end{Bmatrix}$

c. "Unification failure": $\begin{Bmatrix} -\textsc{Back} \\ -\textsc{Round} \\ \underline{+\textsc{High}} \end{Bmatrix} \sqcup \left\{\underline{-\textsc{High}}\right\} \rightsquigarrow \begin{Bmatrix} -\textsc{Back} \\ -\textsc{Round} \\ \underline{+\textsc{High}} \end{Bmatrix}$

[2]Gorman and Reiss (2025b) formalize and exemplify the third operator, the *segment* operator.

For both subtraction and unification rules, the target is a natural class, appearing on the left-hand side of the operator, the change is a feature specification, appearing on the right-hand side of the operator, and environments are optionally specified using natural classes after a forward slash.

(14) Subtraction schema: $[\ldots] \setminus \{\ldots\} \;/\; \ldots$

(15) Unification schema: $[\ldots] \sqcup \{\ldots\} \;/\; \ldots$

Subtraction is used to implement processes like debuccalization (cf. autosegmental delinking; e.g., Benz and Volenec 2023) but there is nothing comparable in classical generative phonology. Unification, in turn, provides a novel solution to specificity. As shown in (13), it is non-vacuous only when the target is underspecified for the feature being added. Thus unification corresponds to *feature-filling* (or *structure-building*) processes. Following much prior work (e.g., Poser 1982, Mester and Itô 1989, Inkelas and Cho 1993, Siptár and Törkenczy 2000), LP decomposes *feature-changing* (or *structure-changing*) processes into two steps: one or more subtraction rules followed by one or more *unification* rules.

We now turn to the question of encoding exchange processes in LP using the notation defined above, giving three attempts.

3.2 A failed attempt

Exchange rule (7), defined using the classical generative phonology notation, is feature-changing because it changes $+\textsc{Voice}$ to $-\textsc{Voice}$ and vice versa. Since LP does not have feature-changing rules of this sort, LP does not have exchange rules. However, LP does have the ability to derive feature-changing processes via a sequence of rules. For example, as a first step towards the Fauxluo exchange, the following rule removes $\pm\textsc{Voice}$ specifications from labial stops followed by a lateral.[3]

(16) $\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \end{bmatrix} \setminus \{\alpha\textsc{Voice}\} \;/\; \_\_ \; [+\textsc{Lateral}]$

However, once this rule applies, the underlying /p, b/ distinction has been lost, and no subsequent rule can refer to it. We thus find ourselves in a similar situation to the ordering paradox (6) above.

3.3 The Duke of York gambit

Bever (1967: 89f.) notes that exchange processes can be generated via what would later be called (e.g., Pullum 1976) a *Duke of York* derivation. Bever's construction uses three

[3]Strictly speaking, the change portion of this rule ought to be written $\{-\textsc{Voice}, +\textsc{Voice}\}$, since LP does not permit rules with a single α (see Bale and Reiss 2018: 454f.). Alternatively, one could express this using two subtraction rules, one removing $-\textsc{Voice}$ and another removing $+\textsc{Voice}$.

feature-changing ordered rules; an LP translation requires a sequence of three unifications and two subtractions. It proceeds as follows. At the start of the derivation, an *ad hoc* feature specification—henceforth +X—is introduced to preserve a trace of the underlying VOICE distinction. Then, this ad hoc specification is used to condition the addition (i.e., unification) and subtraction of VOICE specifications. The ad hoc specification is then removed at the end of the derivation. The first two steps are shown below; note that (18) is repeated from the failed (16).

(17) $\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \\ -\textsc{Voice} \end{bmatrix} \sqcup \{+\mathrm{X}\} \;/\; \_\_ \;[+\textsc{Lateral}]$

(18) $\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \end{bmatrix} \setminus \{\alpha\textsc{Voice}\} \;/\; \_\_ \;[+\textsc{Lateral}]$

The next two rules implement the exchange of VOICE. The first rule is conditioned by the presence of +X. The second is unconditioned, applying vacuously to targets already specified +VOICE, which have already undergone the exchange from −VOICE to +VOICE.

(19) $[+\mathrm{X}] \sqcup \{+\textsc{Voice}\}$

(20) $[\;] \sqcup \{-\textsc{Voice}\}$

A final rule removes the ad hoc specification, which is no longer needed.

(21) $[\;] \setminus \{+\mathrm{X}\}$

This +X specification can correspond to any valued feature, say, +GLOTTAL, that is not otherwise active in the phonology of the language.

3.4 Opportunistic underspecification

A slightly simpler construction makes use of underlying underspecification, a common technique in LP (e.g., Bale et al. 2014; Gorman and Reiss 2025a,b). In Fauxluo, there is a superficial [p] $\leftrightsquigarrow$ [b] exchange of surfacing segments, but it need not be the case that the segments involved are underlyingly fully-specified /p, b/. For concreteness, assume that the final consonant in ‘cactus’ (sg. [nap]) is specified +LABIAL, −SONORANT, −CONTINUANT, −VOICE, and so on, and the final consonant in ‘igloo’ (sg. [nab]) is specified +LABIAL and −SONORANT but is not underlyingly specified for VOICE or CONTINUANT. With this phonemicization, the Fauxluo exchange process can be expressed using four rules rather than the five required for the Duke of York construction.

The first rule adds a −VOICE specification to the root-final consonant of ‘igloo (tr.)’.

(22) $\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \end{bmatrix} \sqcup \{-\textsc{Voice}\} \;/\; \_\_ \;[+\textsc{Lateral}]$

This applies vacuously to 'cactus (tr.)', which is specified $+$VOICE. The second rule removes the $-$VOICE specification from the final consonant of 'cactus (tr.)'.

(23) $\begin{bmatrix} +\textsc{Labial} \\ -\textsc{Sonorant} \\ -\textsc{Continuant} \end{bmatrix} \setminus \{-\textsc{Voice}\} \ / \ \_\_ \ [+\textsc{Lateral}]$

This does not affect 'igloo (tr.)' because the root-final consonant is not $-$CONTINUANT. The final two rules are context-free redundancy rules, applying vacuously in many cases.

(24) $[+\textsc{Labial}] \sqcup \{+\textsc{Voice}\}$

(25) $[\ ] \sqcup \{-\textsc{Continuant}\}$

Abstract sample derivations are provided below, with complete feature specifications (L: LABIAL, C: CONTINUANT, V: VOICE) given for the final segment of each root.

(26) Sample Fauxluo derivations:

	'cactus (sg.)'	'cactus (tr.)'	'igloo (sg.)'	'igloo (tr.)'
UR	+L, −C, −V	+L, −C, −V	+L	+L
Rule (22):				+L, −V
Rule (23):		+L, −C		
Rule (24):		+L, −C, +V	+L, +V	
Rule (25):			+L, −C, +V	+L, −C, −V
SR	+L, −C, −V	+L, −C, +V	+L, −C, +V	+L, −C, −V

3.5 Local summary

In classical generative phonology, the independently-motivated α-notation provides a mechanism to express exchange rules. In LP, exchanges can only be expressed by complex sequences of rules, via the use of either a Duke of York derivation or opportunistic underspecification, but exchanges can still be expressed.

4. Filters and protocols

For a phonological pattern to actually be attested in some language, it must necessarily be statable within whichever model of phonological Universal Grammar (UG) one assumes. For sake of argument, let us assume that exchange processes are in fact unattested, as seems likely. Given then that LP can generate exchange processes, does this mean that the LP model of intrasegmental change is too powerful? Below, we provide a few reasons why one perhaps should not worry about this kind of overgeneration.

The first argument suggests that exchange processes may be victims of the *diachronic filter*, a notion discussed in work by Ohala (2003), Hale (2003), and Blevins (2004), among others. That is, for a statable process to be attested, it must also be possible for it to come into existence via a plausible series of sound changes and/or reanalyses, beginning from plausible initial conditions. Given the apparent abstractness, complexity, and phonetic unnaturalness of the two working LP formulations of exchange processes, it seems plausible that this filter would be responsible for non-attestation. Note that this filter, as explicated by Hale (2007: 140f.), is neither part of UG nor of any particular synchronic grammar: it is merely a way of describing the fact that certain constellations of events involving grammars in descent relationships are highly improbable. If so, then, there is no need to add any complexity to UG itself to account for this apparent typological gap. As Kaplan writes:

> A formal theory may have a relatively smooth outline…[t]hen you start taking chunks out of it…because you claim that no human language or grammar has such and such a property. […] It's a mistake to carry premature and unjustified substantive hypotheses into our computational and mathematical work, especially if it leads to mathematically complex, even if more restrictive, theories. (Kaplan 1995: 346–7)

Not only is it hard to imagine how one might restrict LP to exclude exchange processes, but it is necessary to accept that scientific models generally allow for a wider range of phenomena that what has been observed occurring naturally—this is why chemists are able to produce plastics (Reiss 2022).

Of course, the existence of diachronic improbabilities is not the only filtering condition one might appeal to: perhaps the exchange structure pattern is itself unacquirable. Kiparsky (1968, et. seq.) and Kenstowicz and Kisseberth (1977: ch. 1), among others, propose to limit the apparent abstractness of phonological derivations. The ontological status of these proposals—and similar proposals made during the post-SPE abstractness debate—is not always clear for these authors. For instance, they can be interpreted as negative heuristics for the linguist, or as defeasible inductive biases encoded within the language acquisition device (LAD), or as hard limits on what types of analyses a child can acquire. In lieu of a complete theory of the phonological LAD under LP—our ultimate goal—we propose explicit *LAD protocols* that govern analyses proffered by the LAD. These are intended as contributions to the theory of phonological *acquisition*, not phonological *computation*; they restrict which LP grammars are acquired, not which grammars can be characterized by the LP model. In other words, our LP model intensionally characterizes a set of grammars, but only some of those grammars are acquirable. Any such unacquirable grammar will never be attested under normal circumstances.

For example, recall that the Duke of York analysis (§3.3) depends on an ad hoc feature specification to track segments' underlying VOICE specifications. If the LAD is unable to posit ad hoc feature distinctions that never surface, then this type of Duke of York derivation will not be acquirable. This is stated as a LAD protocol below.

(27) NO WANDERING TARGETS: Any valued feature added to (the correspondent of) a segment by a unification rule must be a member of the generalized union of all valued features found in all allophones of that segment.

This protocol does not rule out all false steps (in the sense of Zwicky 1974) nor all Duke of York derivations, only those which implicate a valued feature which has no correspondents on the surface. We hypothesize that similar protocols might be deployed to rule out the use of opportunistic underspecification (§3.4), but leave this for future work.

5. Conclusions

There is as yet no conclusive evidence for the existence of exchanges, yet exchange *rules* can be generated by the α-notation of classical generative phonology, and LP has various ways to generate exchange *processes*. We propose that their apparent non-existence may reflect either the diachronic filter or non-trivial properties of the phonological LAD. In other words, exchanges may be unacquirable or otherwise unattestable for reasons that have nothing to do with the expressivity of LP grammars. We believe that our proposals about the phonological LAD in particular are an improvement on the vague intuitions about elegance or plausibility commonly found in the literature on phonological abstractness (cf. Zwicky 1974; Pullum 1976), and that they help clarify epistemological and ontological issues—those faced by the linguist versus those faced by the child—often confounded in earlier work. While our discussion focuses on exchanges, we suspect similar logic ought to be applied to understand the apparent non-existence or rarity of other taxonomic categories in phonology, such as metathesis or saltation.

While we have not discussed Optimality Theory (OT) above, the status of exchanges in this theory is also under dispute. Moreton (2004), for example, proves that exchanges are uncomputable in classical OT, but Alderete (2001) proposes to extend the universal constraint set Con with a family of *anti-faithfulness* constraints, which make exchanges computable. Whether or not exchanges exist in OT, then, comes down to whether or not Alderete's anti-faithfulness constraints are admitted to Con.

References

Alderete, John. 2001. Dominance effects as transderivational anti-faithfulness. *Phonology* 18: 201–253.

Anderson, Stephen R., and Wayles Browne. 1973. On keeping exchange rules in Czech. *Papers in Linguistics* 6: 445–482.

Bale, Alan, M. Love, and Charles Reiss. 2025. Variable subsumption and Logical Phonology. https://lingbuzz.net/lingbuzz/009068. Ms., Concordia University.

Bale, Alan, Maxime Papillon, and Charles Reiss. 2014. Targeting underspecified segments: A formal anaylsis of feature-changing and feature-filling rules. *Lingua* 148: 240–253.

Bale, Alan, and Charles Reiss. 2018. *Phonology: A formal introduction*. Cambridge: MIT Press.

Bale, Alan, Charles Reiss, and David Ta-Chun Shen. 2020. Sets, rules and natural classes: { } vs. [ ]. *Loquens* 6: e06.

Benz, Johanna, and Veno Volenec. 2023. Two logical operations underlie all major types of segmental alternations. Paper presented at the 30th Manchester Phonology Meeting.

Bever, Thomas G. 1967. Leonard Bloomfield and the phonology of the Menomini language. Doctoral dissertation, Massachusetts Institute of Technology.

Blevins, Juliette. 2004. *Evolutionary phonology*. Cambridge: Cambridge University Press.

Chomsky, Noam, and Morris Halle. 1965. Some controversial questions in phonological theory. *Journal of*

Linguistics 1: 97–138.

Chomsky, Noam, and Morris Halle. 1968. *The sound pattern of English*. New York: Harper & Row.

Dabbous, Rim, Marjorie Leduc, Charles Reiss, and David Ta-Chun Shen. 2024. Locality is epiphenomenal: Adjacency is opaqueness. In *Actes du congrès annuel de l'association canadienne de linguistique 2024 / Proceedings of the 2024 annual conference of the canadian linguistic association*. https://cla-acl.ca/pdfs/actes-2024/Dabbous-etal-CLA-2024.pdf.

DiCanio, Christian, Basileo Martínez Cruz, Benigno Cruz Martínez, and Wilberto Martínez Cruz. 2020. Glottal toggling in Itunyoso Triqui. *Phonological Data and Analysis* 2(4): 1–28.

Gorman, Kyle, and Charles Reiss. 2025a. Metaphony in Substance Free Logical Phonology. *Phonology* in press.

Gorman, Kyle, and Charles Reiss. 2025b. Natural class reasoning in segment deletion rules. In *Proceedings of the 56th Annual Meeting of the North East Linguistics Society*, in press.

Hale, Mark. 2003. Neogrammarian sound change. In *The handbook of historical linguistics*, ed. Brian D. Joseph and Richard D. Janda, 343–368. Oxford: Blackwell Publishing.

Hale, Mark. 2007. *Historical linguistics: Theory and method*. Malden, MA: Blackwell Publishing.

Inkelas, Sharon, and Young-Mee Yu Cho. 1993. Inalterability as prespecification. *Language* 69(3): 529–574.

Kaplan, Ronald M. 1995. Three seductions of computational psycholinguistics. In *Formal issues in lexical-functional grammar*, ed. Mary Dalrympe, Ronald M. Kaplan, John T. Maxwell, III, and Annie Zaenen, 339–367. Stanford, CA: CSLI Publications.

Kenstowicz, Michael, and Charles Kisseberth. 1977. *Topics in phonological theory*. New York: Academic Press.

Kiparsky, Paul. 1968. *How abstract is phonology?* Bloomington: Indiana University Linguistics Club.

de Lacy, Paul. 2012. Morphophonological polarity. In *Morphology and phonology of exponence*, ed. Jochen Trommer, 121–159. Oxford: Oxford University Press.

de Lacy, Paul. 2020. Do morphophonological exchange rules exist? A reply to DiCanio et al. (2020). *Phonological Data & Analysis* 2(4): 29–43.

Leduc, Marjorie, Charles Reiss, and Veno Volenec. 2024. Votic vowel harmony in Substance Free Logical Phonology. In *Oxford handbook of vowel harmony*, ed. Harry van der Hulst and Nancy A. Ritter, 425–436. New York: Oxford University Press.

Lees, Robert B. 1961. *The phonology of Modern Standard Turkish*. Bloomington: Indiana University Publications.

Lightner, Theodore M. 1971. On Swadesh and Voegelin's 'A Problem in Phonological Alternation'. *International Journal of American Linguistics* 37(4): 227–237.

McCawley, James D. 1974. Review of Chomsky & Halle (1968), *The Sound Pattern of English*. *International Journal of American Linguistics* 40: 50–88.

Meinhof, Carl. 1912. *Die Sprachen der Hamiten*. Hamburg: Friederichsen.

Mester, Armin, and Junko Itô. 1989. Feature predictability and underspecification: Palatal prosody in Japanese mimetics. *Language* 65: 258–293.

Moreton, Elliott. 2004. Non-computable functions in Optimality Theory. In *Optimality theory in phonology: A reader*, ed. John J. McCarthy, 141–164. Malden, MA: Blackwell Publishing.

Ohala, John J. 2003. Phonetics and historical phonology. In *The handbook of historical linguistics*, ed. Brian D. Joseph and Richard D. Janda, 669–686. Oxford: Blackwell Publishing.

Poser, William. 1982. Phonological representation and action at-a-distance. In *The structural of phonological representations*, ed. Harry van der Hulst and Norval Smith, 121–58. Dordrecht: Foris.

Pullum, Geoffrey K. 1976. The Duke of York Gambit. *Journal of Linguistics* 12(1): 83–102.

Reiss, Charles. 2021. Towards a complete Logical Phonology model of intrasegmental changes. *Glossa* 6(1): 107.

Reiss, Charles. 2022. Plastics. In *Ha! Linguistic studies in honor of Mark R. Hale*, ed. Laura Grestenberger Grestenberger, Charles Reiss, Hannes A. Fellner, and Gabriel Z. Pantillon, 327–330. Wiesbaden: Dr. Ludwig Reichert Verlag.

Reiss, Charles. 2024. Specificity in rule targets and triggers: Two v's in Hungarian. In *Actes du congrès annuel* *de l'association canadienne de linguistique 2024 / Proceedings of the 2024 annual conference of the canadian linguistic association*. https://cla-acl.ca/pdfs/actes-2024/Reiss-CLA-2024.pdf.

Reiss, Charles, and Veno Volenec. 2022. Conquer primal fear: Phonological features are innate and substance free. *Canadian Journal of Linguistics* 67(4): 581–610.

Shieber, Stuart. 1986. *An introduction to unification-based approaches to grammar*. Stanford, CA: CSLI Publications.

Siptár, Péter, and Miklós Törkenczy. 2000. *The phonology of Hungarian*. Oxford: Oxford University Press.

Tucker, Archibald N. 1994. *A grammar of Kenya Luo (Dholuo)*. Cologne: Rüdiger Köppe Verlag.

Wolfe, Patricia M. 1970. Some theoretical questions on the historical English vowel shift. *Papers in Linguistics* 3: 221–235.

Zonneveld, Wim. 1976. An exchange rule in Flemish Brussels Dutch. *Linguistic Analysis* 2(2): 109–114.

Zwicky, Arnold M. 1974. Taking a false step. *Language* 50(2): 215–224.